

ECE 3337: Test #2

Spring 2025

Last name of student POSTED SOLUTIONS	Student ID number _____
First name of student _____	Email address _____

Do This First:

1. Make sure that you have all the pages of the exam.
2. Fill in the above information.
3. Turn off and put away all electronic gadgets (laptop, tablet, smartphone, calculator, smart watch, ...)

Exam Rules:

1. You are allowed to bring one 8.5" x 11" handwritten crib sheet (double sided).
2. Indicate your answers in the spaces provided. Use the back of the page to carry out calculations.

Q1	Q2	Q3	Q4	Q5	Q6	Total
20	20	25	15	10	10	100
Laplace Transform	Inverse Laplace Transform	Circuit Analysis	Response Partitions	Op Amp Circuits	Block Diagrams	_____
_____	_____	_____	_____	_____	_____	_____

1. (20 points) Laplace Transforms

a. (6 points) Find $X(s)$, the Laplace transform of the following signal:

$$x(t) = 2\delta(3t - 5) + (t - 1)^2 e^{-2t} u(t - 1) + t \sin(t) u(t).$$

$$X(s) = \frac{2}{3} e^{-\frac{5s}{3}} + 2 \frac{e^{-(s+2)}}{(s+2)^3} + \frac{2s}{(1+s^2)^2}$$

(A) (B) (C)

Grading:

- 1 pt minimum for attempt
- 2 pts each (A), (B), (C) (max 6 pts)

Details:

$$\textcircled{A}: \delta(t) \rightarrow \delta(3t) \rightarrow \delta\left(3\left(t - \frac{5}{3}\right)\right) \rightarrow 2\delta\left(3\left(t - \frac{5}{3}\right)\right)$$

$$1 \rightarrow \frac{1}{3} \cdot 1 \rightarrow e^{-\frac{5s}{3}} \cdot \frac{1}{3} \rightarrow \frac{2}{3} e^{-\frac{5s}{3}}$$

$$\textcircled{B}: t^2 e^{-2t} u(t) \rightarrow (t-1)^2 e^{-2(t-1)} u(t-1) \rightarrow e^{-2} \cdot (t-1)^2 e^{-2(t-1)} u(t-1)$$

$$\left(\frac{2}{(s+2)^3}\right) \rightarrow e^{-s} \cdot \frac{2}{(s+2)^3} \rightarrow e^{-2} \cdot e^{-s} \cdot \frac{2}{(s+2)^3} = 2 \cdot \frac{e^{-(s+2)}}{(s+2)^3}$$

$$\textcircled{C}: \sin(t) u(t) \rightarrow t \sin(t) u(t)$$

$$\frac{1}{s^2+1} \rightarrow -\frac{d}{ds} \left\{ \frac{1}{s^2+1} \right\} = \frac{2s}{(s^2+1)^2}$$

(b) (6 points) Given a signal $x_2(t)$, find the Laplace transform of $y(t) = 5e^{-t} \int_{0-}^t x_2(2\tau) d\tau$ in terms of $X_2(s)$

$$x_2(t) \rightarrow x_2(2t) \rightarrow \int_{0-}^t x_2(2\tau) d\tau \rightarrow e^{-t} \int_{0-}^t x_2(2\tau) d\tau \rightarrow 5e^{-t} \int_{0-}^t x_2(2\tau) d\tau$$

$$X_2(s) \rightarrow \frac{1}{2} X_2\left(\frac{s}{2}\right) \rightarrow \frac{1}{2s} X_2\left(\frac{s}{2}\right) \rightarrow \frac{1}{2(s+1)} X_2\left(\frac{s+1}{2}\right) \rightarrow \frac{5}{2(s+1)} X_2\left(\frac{s+1}{2}\right)$$

(A) (B) (C) (D) Answer (E)

Grading:

- 1 pt for attempt
- +5 pt correct (E)
- 1 pt partial credit for steps (A), (B), (C), (D) being correctly done.

c. (8 points) Find the Laplace transform of $z(t) = t e^{-t} \sin(2t) u(t)$, and find the values of all its poles and zeros.

$$e^{-t} \sin(2t) u(t) \rightarrow t e^{-t} \sin(2t) u(t) \quad \textcircled{A}$$

$$\frac{2}{(s+1)^2 + 2^2} \rightarrow -\frac{d}{ds} \left\{ \frac{2}{(s+1)^2 + 4} \right\} \quad \textcircled{B}$$

$$= \frac{4(s+1)}{((s+1)^2 + 4)^2} \quad \textcircled{C}$$

Grading:

- 2 pts for attempt
- +2 pts each for $\textcircled{A}$, $\textcircled{B}$, $\textcircled{C}$

2. (20 points) Inverse Laplace Transform

Find the inverse Laplace transforms for the following systems.

a. (10 points) $X(s) = \frac{(s+5)}{(s^2+4s+13)} e^{-3s}$ (A) (B) (C)

Let $X_1(s) = \frac{(s+5)}{(s^2+4s+13)} = \frac{(s+2+3)}{(s^2+2.2s+2^2+9)} = \frac{(s+2)+3}{(s+2)^2+3^2} = \frac{(s+2)}{(s+2)^2+3^2} + \frac{3}{(s+2)^2+3^2}$

The inverse Laplace transform of each of the terms is

$\frac{(s+2)}{(s+2)^2+3^2} \xleftrightarrow{\mathcal{L}} e^{-2t} \cos(3t) u(t)$ and $\frac{3}{(s+2)^2+3^2} \xleftrightarrow{\mathcal{L}} e^{-2t} \sin(3t) u(t)$ (D) (E)

Combining, $X_1(s) \xleftrightarrow{\mathcal{L}} x_1(t) = e^{-2t} \cos(3t) u(t) + e^{-2t} \sin(3t) u(t)$

Using the time shifting property to include the effect of e^{-3s} , (F)

$x(t) = e^{-2(t-3)} \cos(3(t-3)) u(t-3) + e^{-2(t-3)} \sin(3(t-3)) u(t-3)$. (G) (H)

2 points for trying. 1 point for A-H.

Alternate form: The solution can be in the form of complex numbers instead of sin and cosine terms:

$$x_1(t) = \frac{1}{2} e^{-2t} [(1-j)e^{j3t} + (1+j)e^{-j3t}] u(t)$$

$$x(t) = \frac{1}{2} e^{-2(t-1)} [(1-j)e^{j3(t-1)} + (1+j)e^{-j3(t-1)}] u(t-1)$$

b. (10 points) $X(s) = \frac{s^2+3s+8}{s^2+6s+8}$

The order of the numerator is equal to the order of the denominator. Therefore, perform long division.

$$X(s) = \frac{s^2+3s+8}{s^2+6s+8} = \frac{s^2+6s-3s+8}{s^2+6s+8} = 1 - \frac{3s}{s^2+6s+8} = 1 - \frac{3s}{(s+2)(s+4)}$$
 (A) (B) (C)

Separating into Partial fractions:

$$X(s) = 1 - \frac{3s}{(s+2)(s+4)} = 1 - \left[\frac{A}{(s+2)} + \frac{B}{(s+4)} \right]$$

Solving, $A = -3, B = 6$

(D) (E)

$$X(s) = 1 + \frac{3}{(s+2)} - \frac{6}{(s+4)}$$

Finding the inverse of each term gives

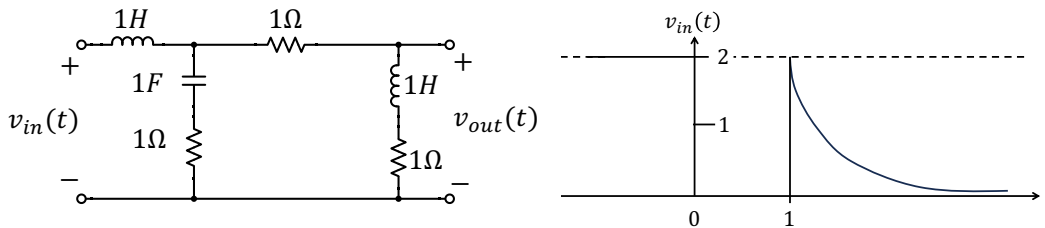
$$x(t) = \delta(t) + 3e^{-2t}u(t) - 6e^{-4t}u(t)$$

2 points for trying. 1 point for A-H.

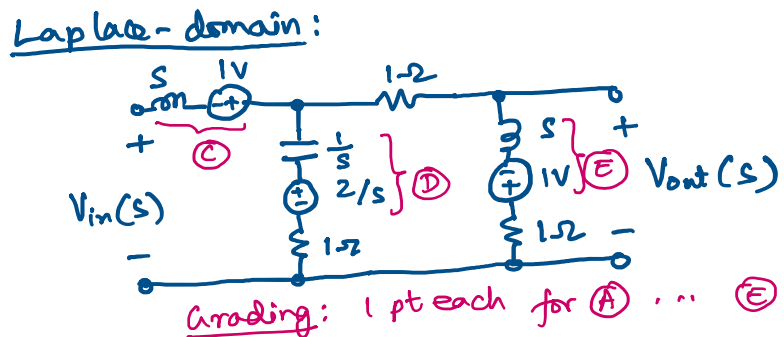
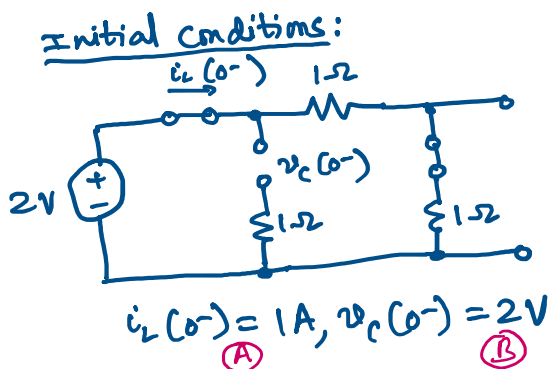
(F) (G) (H)

3. (25 points) Circuit Analysis

For the circuit below, the input $V_{in}(t) = x(t)$ is given in the graph. The units for $x(t)$ is Volts [V].



- a. (5 points) Draw the s-domain representation of the circuit with the initial conditions indicated.



- b. (5 points) Find the s-domain representation of the input $v_{in}(t)$ for $t > 0$ if the coefficient of the exponential is 1.

$$v_{in}(t) = e^{-(t-1)} u(t-1)$$

$$\Rightarrow V_{in}(s) = e^{-s} \cdot \frac{1}{(s+1)}$$

Grading:

- 1 pt for attempt
- + 4pts correct answer

- c. (10 points) Write the mesh current equations in the s-domain needed to find the output $V_{out}(s)$. Express your answer by filling in the entries in the Cramer's method matrices shown below. Here, I_1 and I_2 represent the mesh currents (left to right), and V_1 , V_2 represent mesh voltages.

$$\begin{pmatrix} A_1 & A_2 \\ A_3 & A_4 \end{pmatrix} \begin{pmatrix} I_1 \\ I_2 \end{pmatrix} = \begin{pmatrix} V_1 \\ V_2 \end{pmatrix}$$

$$\begin{aligned}
 I_1 \left(s + \frac{1}{s} + 1 \right) - I_2 \left(\frac{1}{s} + 1 \right) &= V_{in}(s) - \frac{2}{s} \\
 -I_1 \left(\frac{1}{s} + 1 \right) + I_2 \left(1 + \frac{1}{s} + 1 + s + 1 \right) &= \frac{2}{s} + 1
 \end{aligned}$$

Grading:
 • 2 pts attempt
 + 2 pts each
 $A_1, A_2, A_3, A_4,$
 $V_1, V_2.$

$$\Delta = A_1 A_4 - A_3 A_2 = \frac{s^3 + 4s^2 + 4s + 2}{s}$$

$$\Delta I_2 = A_1 V_2 - A_3 V_1 = \frac{s^2 + 3s + 1 + (s+1) V_{in}(s)}{s}$$

$$\Rightarrow I_2 = \frac{\Delta I_2}{\Delta} = \frac{s^2 + 3s + 1 + (s+1) V_{in}(s)}{s^3 + 4s^2 + 4s + 2}$$

$$= \frac{s^2 + 3s + e^{-s} V_{in}}{s^3 + 4s^2 + 4s + 2} \quad (A)$$

d. (5 points) Write the formula for $V_{out}(s)$ in terms of $V_{in}(s)$. Your answer should be written as a ratio of polynomials.

$$V_{out}(s) = (3s + 1) I_2(s) + 1 \quad (B)$$

$$= (3s + 1) \frac{(s^2 + 3s + e^{-s} V_{in})}{s^3 + 4s^2 + 4s + 2} + 1$$

Grading:
 • 1 pt for attempt
 • +2 pts for I_2 (A)
 • +2 pts for V_{out} (B)

4. (15 points) System Response Partitions

A linear time-invariant system is described by the following LCCDE

$$\frac{d^2 y(t)}{dt^2} + 5 \frac{dy(t)}{dt} + 6y(t) = x(t)$$

The input is $x(t) = \sin(t) u(t)$, with the initial conditions $y(0^-) = 0$, $y'(0^-) = 1$.

- a. (5 points) Derive a formula for the output $Y(s)$. Your answer should be expressed as a ratio of polynomials with a fully factored denominator.

$$\begin{aligned} y(t) &\stackrel{\mathcal{L}}{\leftrightarrow} Y(s) \\ \frac{d}{dt} y(t) &\stackrel{\mathcal{L}}{\leftrightarrow} sY(s) - y(0^-) \\ \frac{d^2}{dt^2} y(t) &\stackrel{\mathcal{L}}{\leftrightarrow} s^2 Y(s) - sy(0^-) - y'(0^-) \end{aligned}$$

Therefore, taking the Laplace transform of the LCCDE yields:

$$\begin{aligned} s^2 Y(s) - sy(0^-) - y'(0^-) + 5(sY(s) - y(0^-)) + 6Y(s) &= X(s) \\ s^2 Y(s) + 5sY(s) + 6Y(s) - sy(0^-) - y'(0^-) - 5y(0^-) &= X(s) \end{aligned} \quad \text{--- (A)}$$

Using the given initial conditions

$$\begin{aligned} s^2 Y(s) + 5sY(s) + 6Y(s) - 1 &= X(s) \\ s^2 Y(s) + 5sY(s) + 6Y(s) &= X(s) + 1 \end{aligned}$$

$$Y(s)(s^2 + 5s + 6) = X(s) + 1 \quad \text{--- (B)}$$

Also,

$$\begin{aligned} x(t) &\stackrel{\mathcal{L}}{\leftrightarrow} X(s) \\ \sin(t) u(t) &\stackrel{\mathcal{L}}{\leftrightarrow} \frac{1}{s^2 + 1} \end{aligned}$$

Substituting $X(s) = \frac{1}{s^2 + 1}$, (C)

$$Y(s)(s^2 + 5s + 6) = \frac{1}{s^2 + 1} + 1 = \frac{1 + s^2 + 1}{(s^2 + 1)} = \frac{s^2 + 2}{(s^2 + 1)}$$

$$Y(s) = \frac{s^2 + 2}{(s^2 + 1)(s^2 + 5s + 6)} = \frac{s^2 + 2}{(s^2 + 1)(s + 3)(s + 2)}$$

$$Y(s) = \frac{s^2 + 2}{(s^2 + 1)(s + 3)(s + 2)} \quad \text{--- (D)}$$

Alternate form: $Y(s) = \frac{s^2 + 2}{(s + j)(s - j)(s + 3)(s + 2)}$

1 point for trying. 1 point for A-D.

- b. (5 points) Find the modes of the system represented by the above LCCDE. Calculate the natural response of this system in the Laplace domain.

The modes of the system correspond to the roots of the characteristic equation. The characteristic equation is: $s^2 + 5s + 6 = 0$ or $(s + 3)(s + 2) = 0$ or $s = -3, -2$

Calculating the partial fraction of

$$Y(s) = \frac{s^2 + 2}{(s^2 + 1)(s + 3)(s + 2)} = \frac{1}{10} \cdot \frac{-s + 1}{s^2 + 1} + \frac{6}{5(s + 2)} - \frac{11}{10} \frac{1}{(s + 3)}$$

The natural response corresponds to the terms

$$Y_{\text{natural}}(s) = \frac{6}{5(s + 2)} - \frac{11}{10} \frac{1}{(s + 3)}$$

1 point for trying. 1 point for A-D.

- c. (5 points) Calculate the initial value $y(0+)$ of this system's response.

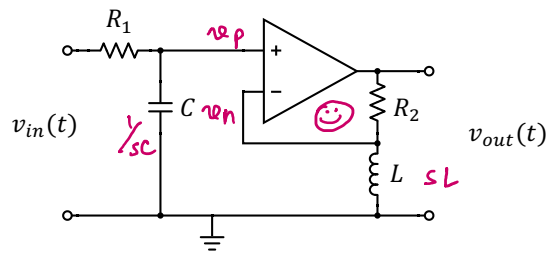
To find the initial value, we can use the initial value theorem.

$$y(0+) = \lim_{s \rightarrow \infty} sY(s) = \lim_{s \rightarrow 0} s \frac{s^2 + 2}{(s^2 + 1)(s + 3)(s + 2)} = 0$$

The limit is 0 since the numerator goes to 0, but the denominator does not go to 0.

1 point for trying. 1 point for A-D.

5. (10 points) Operational Amplifiers



a. (6 points) Calculate the transfer function $\mathbf{H(s)}$ for this circuit.

$$V_p(s) = \frac{1/sC}{R_1 + 1/sC} \cdot V_{in}(s) = V_n(s) = \frac{sL}{R_2 + sL} \cdot V_{out}(s)$$

$$\Rightarrow \frac{V_{out}(s)}{V_{in}(s)} = \frac{(R_2 + sL) \overset{\text{zero}}{\uparrow}}{sL(1 + sCR_1) \overset{\text{pole}}{\uparrow} \overset{\text{pole}}{\uparrow}}$$

Grading:

- 2 pts for attempt
- + 4 pts correct answer

b. (4 points) Identify the poles and zeros of the transfer function $\mathbf{H(s)}$.

poles @ $\left\{ 0, -\frac{1}{CR_1} \right\}$

(A) (B)

zero @ $\left(-\frac{R_2}{L} \right)$

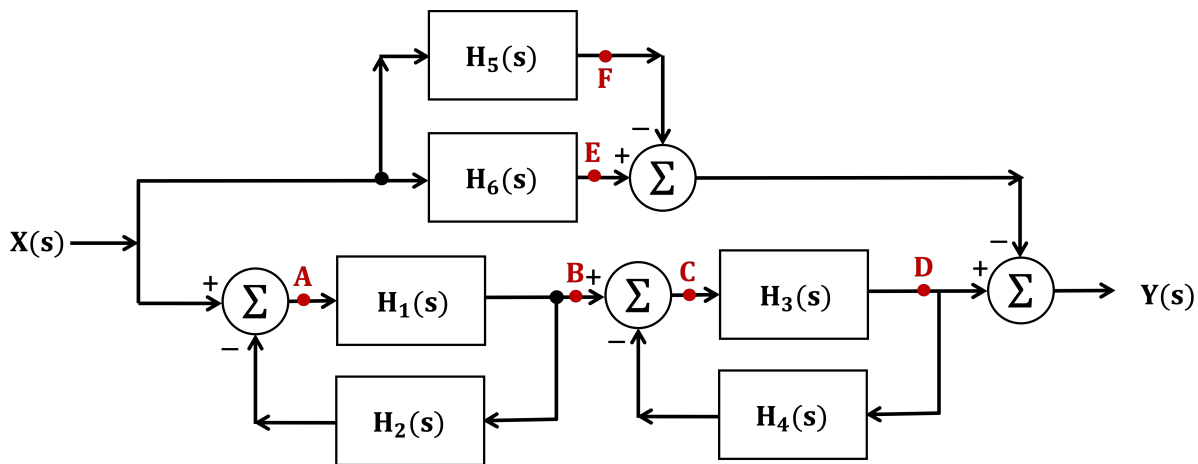
(C)

Grading:

- 1 pt attempt
 - + 1 pt each pole & zero
- (A), (B), (C)

6. (10 points) System Block Diagrams

Calculate the overall transfer function $\mathbf{H(s)}$ for the following connection of LTI systems. Your final answer should be expressed in the Laplace domain.



$$A = X - BH_2$$

$$B = AH_1$$

Substituting for A and simplifying:

$$B = (X - BH_2)H_1$$

$$\text{Or } B = X \frac{H_1}{1 + H_1 H_2}$$

Similarly

$$C = B - DH_4$$

$$D = CH_3$$

Substituting for C and simplifying:

$$D = (B - DH_4)H_3$$

$$\text{Or } D = B \frac{H_3}{1 + H_3 H_4} = X \frac{H_1}{1 + H_1 H_2} \cdot \frac{H_3}{1 + H_3 H_4}$$

The top part yields

$$F = XH_5$$

$$E = XH_6$$

$$Y = D - (E - F) = D - X(H_6 - H_5)$$

$$Y = X \frac{H_1}{1 + H_1 H_2} \cdot \frac{H_3}{1 + H_3 H_4} - X(H_6 - H_5)$$

The transfer function is

$$H = \frac{H_1}{1 + H_1 H_2} \cdot \frac{H_3}{1 + H_3 H_4} - (H_6 - H_5)$$

2 points for trying. 1 point for A-H.